C - Vanish

分值：300300300 分

Problem Statement

You are given an integer sequence $A = (A1, A2, \dots, AN) A = (A_1, A_2, \dots, A_N) A = (A1, A2, \dots, AN)$.

给定一个整数序列 $A = (A1, A2, \dots, AN) A = (A_1, A_2, \dots, A_N) A = (A1, A2, \dots, AN)$。

Find the minimum possible sum of all elements of AAA after performing the following operation exactly KKK times.

求恰好执行 KKK 次下述操作后，AAA 的所有元素之和的最小可能值。

- Choose an integer xxx. For each iii such that $Ai = xA_i = xAi = x$, replace the value of AiA_iAi with 000.
- 选择一个整数 xxx。对于每个满足 $Ai = xA_i = xAi = x$ 的 iii，将 AiA_iAi 的值替换为 000。

Constraints

- $1 \leq K \leq N \leq 3 \times 1051 \leq K \leq N \leq 3 \times 10^5 1 \leq K \leq N \leq 3 \times 105$
- $1 \leq Ai \leq 1091 \leq A_i \leq 10^9 1 \leq Ai \leq 109$
- All input values are integers.
- $1 \leq K \leq N \leq 3 \times 1051 \leq K \leq N \leq 3 \times 10^5 1 \leq K \leq N \leq 3 \times 105$
- $1 \leq Ai \leq 1091 \leq A_i \leq 10^9 1 \leq Ai \leq 109$
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入按照以下格式从标准输入给出：

$N N N\ K K K$
$A1 A_1 A1\ A2 A_2 A2 \ldots\ldots\ldots\ AN A_N AN$

Output

Output the answer.

输出答案。

Sample Input 1

```
6 2
7 2 7 2 2 9
```

Sample Output 1

```
6
```

Initially, $A=(7,2,7,2,2,9)A=(7,2,7,2,2,9)A=(7,2,7,2,2,9)$.

初始时，$A=(7,2,7,2,2,9)A=(7,2,7,2,2,9)A=(7,2,7,2,2,9)$。

Performing the operation with $x=9x=9x=9$ gives $A=(7,2,7,2,2,0)A=(7,2,7,2,2,0)A=(7,2,7,2,2,0)$.

选择 $x=9x=9x=9$ 执行一次操作后，得到 $A=(7,2,7,2,2,0)A=(7,2,7,2,2,0)A=(7,2,7,2,2,0)$。

Next, performing the operation with $x=7x=7x=7$ gives $A=(0,2,0,2,2,0)A=(0,2,0,2,2,0)A=(0,2,0,2,2,0)$.

接下来，选择 $x=7x=7x=7$ 执行一次操作后，得到 $A=(0,2,0,2,2,0)A=(0,2,0,2,2,0)A=(0,2,0,2,2,0)$。

At this point, the sum of all elements of AAA is $0+2+0+2+2+0=60+2+0+2+2+0=60+2+0+2+2+0=6$.

此时，AAA 的所有元素之和为 $0+2+0+2+2+0=60+2+0+2+2+0=60+2+0+2+2+0=6$。

Sample Input 2

```
8 6
1 2 3 4 1 2 3 4
```

Sample Output 2

```
0
```

Sample Input 3

```
10 2
3 3 4 1 1 3 3 1 5 1
```

Sample Output 3

```
8
```